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Exacerbation of antenatal care inequities as coverage recommendations rise in Brazil: evidence for the inverse equity hypothesis from national health information system data (SINASC) in 2023

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Abstract

Background Within-country inequalities in antenatal care (ANC) coverage are widely documented. With recent international guidelines recommending eight or more visits, we hypothesized that this increase could reveal wider inequalities in Brazil, consistent with the Inverse Equity Hypothesis, which postulates that new interventions initially widen socioeconomic gaps because privileged groups are the first to adopt them. We assess how the magnitude of inequality varies by increasing ANC visit thresholds.

Methods We analyzed data from the Brazilian Live Births Information System (SINASC) for 2,537,511 live births in 2023, calculating national coverage 1 + to 8 + ANC visits. Estimates were stratified by maternal schooling, self-reported race/skin color, age at birth, and geographic region. Measures of inequality included the Slope Index of Inequality (SII) and the mean absolute difference to the overall mean (MADM).

Results National coverage for 1 + ANC visits was 99.4% and for 8 + visits, 67.4%. For maternal schooling, SII rose from 1.6% points (p.p.) for 1 + visit to 26.4 p.p. for 8 + visits. For 8 + visits, indigenous women showed the largest dropout rates and the lowest coverage (37.8%), whereas white women showed the opposite (74.7% coverage). The North region and adolescents also showed low coverage of 8 + ANC visits (50.8% and 55.4%, respectively).

Conclusions Increasing the number of recommended ANC visits exacerbates socioeconomic, racial, and regional inequities in Brazil, consistent with the inverse equity hypothesis. Higher coverage thresholds disclose wider inequities, thus further exposing the vulnerability of indigenous, with no formal schooling, adolescent women and residents of the North. Improved detection of such inequities is essential to guide policies that ensure high-quality ANC reaches all women and children, thereby eliminating current inequalities in maternal and newborn health outcomes.

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Keywords Prenatal care, Health inequities, Socioeconomic factors, Brazil, Health program implementation

Background

Within-country inequalities in antenatal care (ANC) coverage have been widely studied through population-based studies, which rely on stratification by socioeconomic position and women's schooling, age, and race or ethnicity [1–4]. Subnational regional inequities have also been documented in many countries [5, 6]. The Countdown to 2030 initiative disseminates country equity profiles for 122 low- and middle-income countries (LMICs) using data from national surveys analyzed at the International Center for Equity in Health at the Federal University of Pelotas in Brazil (www.equidade.org). The profiles (available at https://www.countdown2030.org/equity-profiles) display the proportions of pregnant women who have had four or more antenatal care visits, categorized by wealth quintiles, urban/rural residence, subnational regions, and women's educational attainment. The cutoff of four visits reflects the 2002 World Health Organization (WHO) recommendation [7], which was updated in 2016 to eight or more visits [8].

Other antenatal care indicators have also been proposed to report on the number of attendances. One or more visits with an appropriate health care provider (defined according to national criteria) have been widely used for global monitoring in the past (https://data.unicef.org/topic/maternal-health/antenatal-care/). Our study focuses on Brazil, where six visits were recommended in 2000 [9] and seven in 2024 [10].

We use the inverse equity hypothesis framework to explore how changing recommendations on the number of visits may affect the study of health inequalities. Building up on Julian Tudor Hart's 1971 inverse care law (which states that "the availability of good medical or social care tends to vary inversely with the need of the population served" [11]), the inverse equity hypothesis posits that new health interventions will tend to initially reach the most affluent sectors of society, thereby potentially increasing inequality in coverage and health outcomes [12].

Using Brazilian data, we explore how ANC coverage and inequalities vary according to the number of recommended visits indicator. Our motivation was to explore how such changes might affect the magnitude of socioeconomic inequalities in antenatal care.

Methods

The present cross-sectional analyses relied upon Brazilian data obtained from the Live Births Information System (Sistema de Informação sobre Nascidos Vivos, SINASC) for the year 2023. SINASC is based on the Live Birth Declaration (Declaração de Nascido Vivo, DNV), which is expected to be completed for every live birth in the country. The declaration, with national coverage estimated at 99.6% [13, 14], includes information on pregnancy and birth, such as the total number of antenatal care visits [15].

Eight different ANC coverage indicators were studied, representing the proportion of pregnant women who attended from one to eight visits. ANC estimates were presented for four stratifiers: schooling (number of completed years of study categorized as none, 0–3, 4–7, 8–11, ≥ 12 years), self-reported maternal race or skin color (using the official Brazilian classification [16]: white, black, brown, indigenous and excluding the small category of Asian descendants reporting yellow skin color), women's age at birth (< 20, 20–24, 25–29, 30–34, >=35 years), and geographic region of residence (North, Northeast, Southeast, South, Center-West). Equiplots (https://equidade.org/equiplot) were presented to visually show inequalities. We also calculated the ANC dropout rate, defined as the difference between the proportions of women who attended one or more visits and those with eight or more visits, expressed as percentage points (p.p.).

Two summary measures of inequalities were calculated: the slope index of inequality (SII) and the mean absolute difference to the overall mean (MADM). The SII is a measure of absolute inequality for ordered groups, which was based on a logistic regression of ANC visits on schooling and women's age [17]. It is interpretable as the difference in percent points (p.p.) in ANC coverage between women at the top and bottom of the schooling and age distributions, with positive values indicating that coverage increases with wealth or age. The MADM was selected to assess inequality across unordered groups and is defined as $1/N\ \Sigma(yi-\bar{y})$ where y_i represents the ANC coverage, $\bar{y}$ is the mean coverage in each race/skin color group or region, and N is the number of units considered [18]. The MADM is also expressed in p.p. and quantifies the average distance between the coverage in each race/skin color and regional subgroups and the overall population mean, with higher values indicating greater inequalities. Individual data were used in the analyses, and the only exclusion criterion was missing data. All statistical analyses and graphs were done with Stata (StataCorp. 2023. Stata Statistical Software: Release 18. College Station, TX: StataCorp LLC.). The SINASC data is publicly available, provided by the Ministry of Health through its online portal (https://datasus.saude.gov.br/transferencia-de-arquivos/). All information on mothers and live births is anonymized, and the analyses do not require approval by a local ethics committee.

Results

A total of 2,537,511 live births were registered in the SINASC during 2023. The numbers of births with valid information on the women's schooling, race/skin color, age, and region of the country were, respectively, 2,520,414 (99.3%), 2,473,079 (97.5%), 2,537,480 (99.9%), and 2,537,511 (100%). Missing values were excluded, and the samples for each analysis included all births with valid information for the corresponding stratification variable.

National coverage with one or more ANC visits was 99.4% and declined with increasing numbers of visits reaching 67.4% for eight or more visits (Supplementary Table 1).

Inequalities in ANC according to the women's schooling are presented in an equiplot (Fig. 1) where each dot represents the coverage by groups of years of study. The magnitude of inequalities increases with the number of ANC visits, but regardless of the number of visits, we observe direct associations between coverage and schooling. Additional results are presented in Supplementary Table 1, showing that SIIs increased from 1.6 p.p. for 1 + visits to 26.4 p.p. for 8 + visits, indicating the coverage difference among women on the top and bottom of the schooling distribution. Coverage among women with no schooling tends to be considerably lower than among all other groups, indicating a bottom-inequality or marginal exclusion pattern. To examine dropouts, we subtracted coverage with 8 + visits from coverage with 1 + visits; this ranged from 57.1 p.p. for women with no schooling to 23.2 p.p. for those with 12 or more years (Supplementary Table 1).

ANC inequalities according to the women's race/skin color are shown in Fig. 2 and Supplementary Table 2. Except for 1 + visits, where all groups virtually reached universal coverage, indigenous women showed the lowest coverage levels (37.8% for 8 + visits), and white women the highest coverage (74.7% for 8 + visits). Black and brown women showed nearly identical coverage levels, regardless of the number of visits. As observed for schooling, there is a clear pattern of increasing inequality for higher numbers of ANC visits for all groups. The MADM increases monotonically with the number of visits (Fig. 2). Dropout from 1 + to 8 + visits ranged from 24.9 p.p. for white women to 59.9 p.p. for indigenous women,

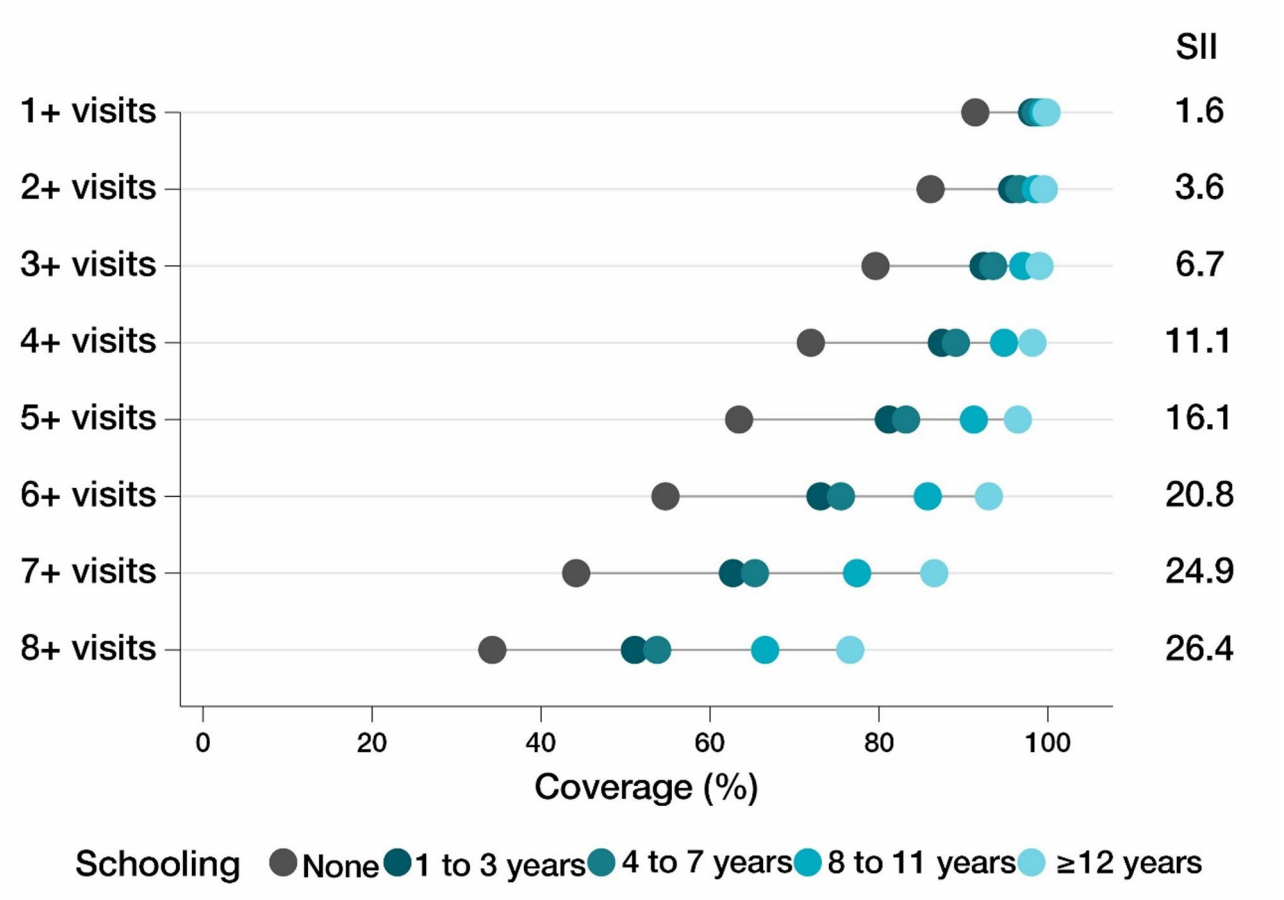

Fig. 1 Inequalities in the number of antenatal care visits according to women's schooling. SII: slope index of inequality in percentage points

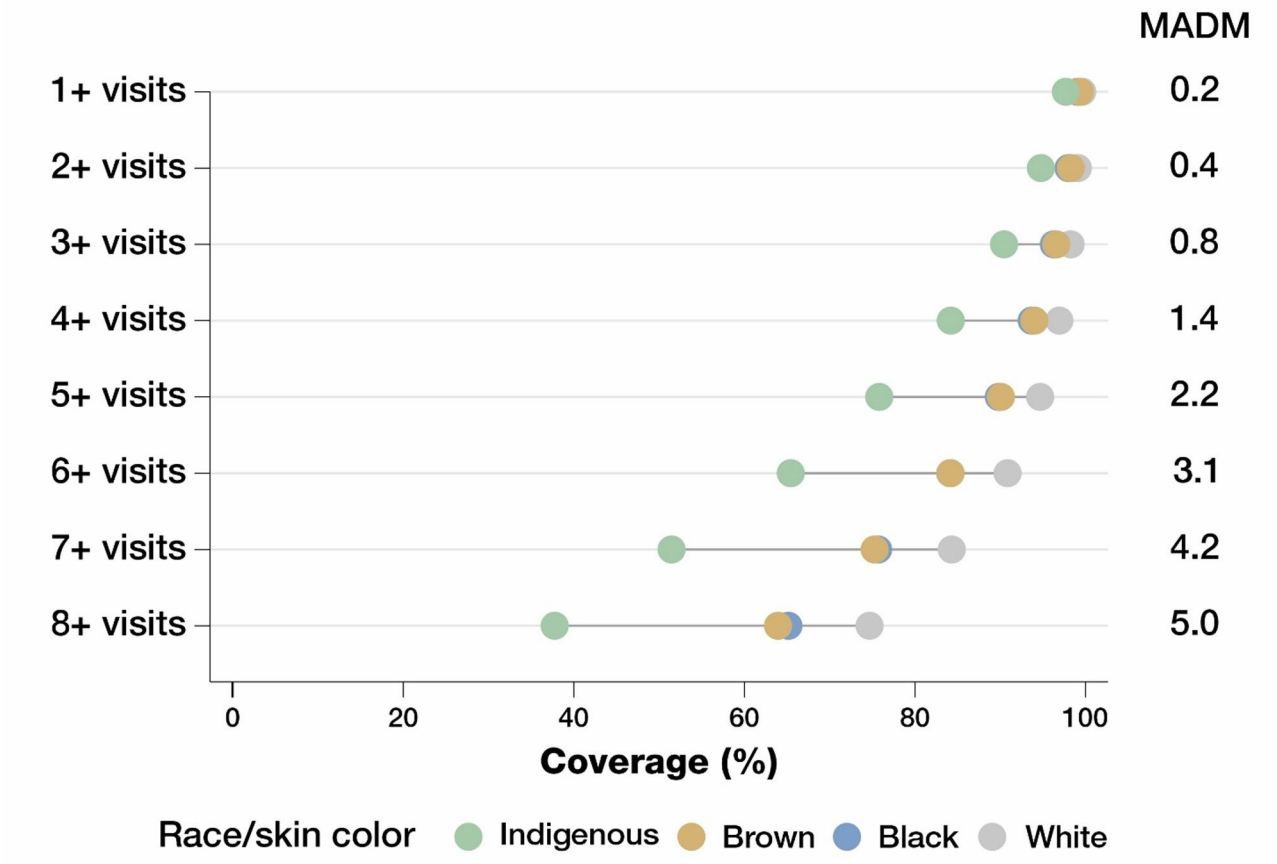

Fig. 2 Inequalities in the number of antenatal care visits according to women's race/skin color. MADM: mean absolute difference to the overall mean in percentage points

black and brown women presented a dropout of 33.9 p.p. and 35.4 p.p., respectively (Supplementary Table 2).

Inequalities according to the geographic region of residence are presented in Fig. 3 and Supplementary Table 3. Coverage tended to be lowest in the North and highest in the South region. Coverage is above 90% in all regions for 1+, 2 + and 3 + visits. For 4 + visits, coverage is below 90% only in the North region, being above 95% in all the other regions. Again, the magnitude of inequality increased with the number of visits, with MADM values increasing from 0.1 p.p. for 1 + visits to 5.6 p.p. for 8 + visits. Women in the North region lag behind in terms of ANC coverage, especially for 4 + visits or more (Fig. 3). Dropout from 1 + to 8 + visits ranged from 23.5 p.p. in the South to 47.9 p.p. in the North (Supplementary Table 3).

Women's age at delivery was also associated with ANC inequalities (Fig. 4 and Supplementary Table 4). Up to 4 + visits, the magnitude of inequalities was small, but for larger numbers of visits, adolescents had lower coverage, while the highest levels were for women aged 35 or more years. Values of the SII increased monotonically, from 0.4 p.p. for 1 + visits to 18.1 p.p. for 8 + visits (Supplementary Table 4). Dropout from 1 + to 8 + visits ranged from 43.7 p.p. in adolescents to 26.9 p.p. among the oldest group (Supplementary Table 4).

We also investigated schooling-related inequalities by race/skin color (Fig. 5). Women's schooling was positively associated with the number of antenatal visits within every racial category. As observed in the previous analyses, the magnitude of inequality increased with the number of visits.

The narrowest schooling gaps were among white women (with SIIs ranging from 0.9 p.p. for 1 + visits to 16.9 p.p. for 8 + visits) and the widest gaps for indigenous women (SIIs of 9.4 p.p. to 41.7 p.p., respectively). For black and brown women, the magnitude of educational inequality as measured by the SII was around 1.5 and 2.4 higher than observed for white women. Fewer than 12% of indigenous women without schooling had 8 + visits, compared to 80% of white women with 12 or more years. Even for coverage of 1 + ANC visits, for which the national level was high, education-based inequalities were over 10 times wider for indigenous than for white women. By examining Fig. 5 vertically, it becomes visually evident that racial inequalities follow a clear gradient, with coverage decreasing from white to black, brown,

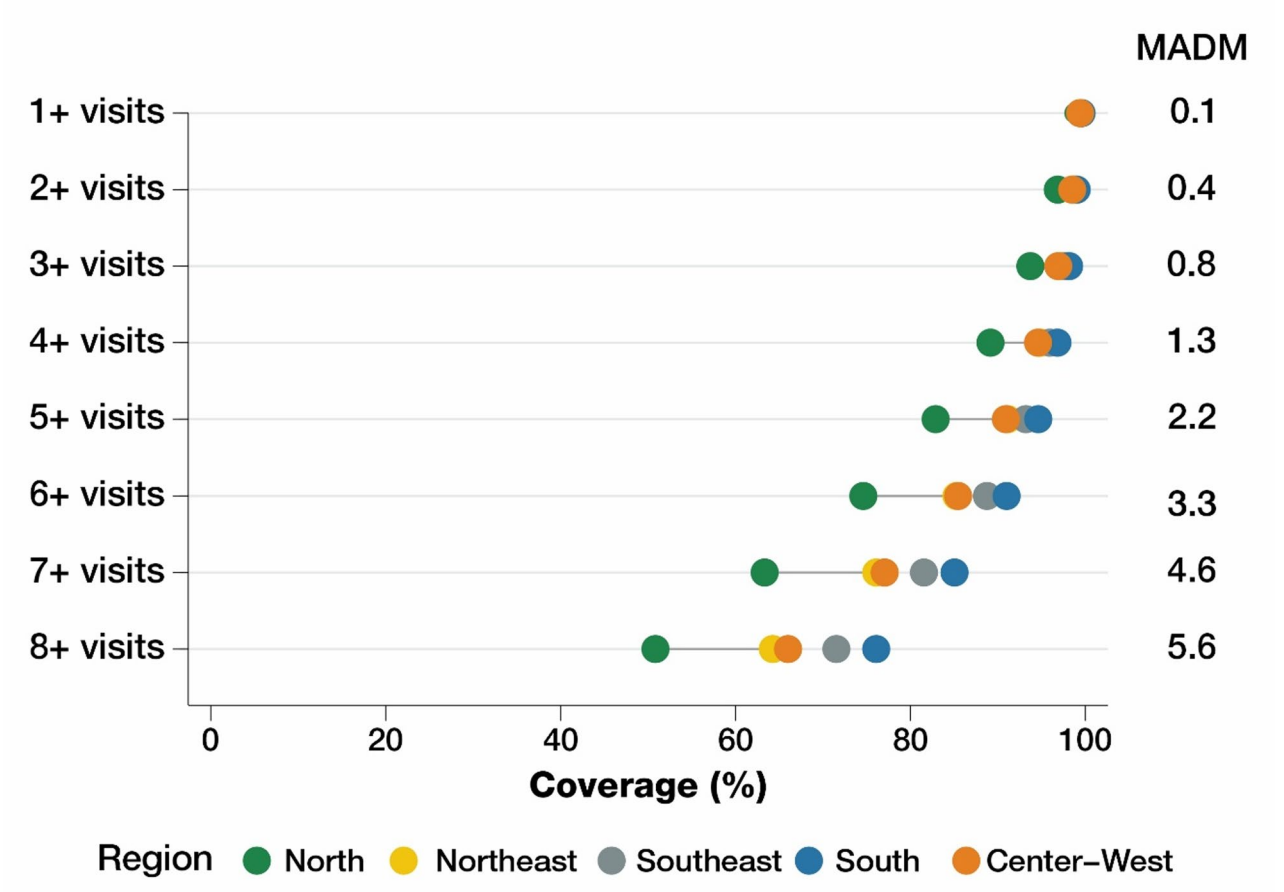

Fig. 3 Inequalities in the number of antenatal care visits according to the region of the country. MADM: mean absolute difference to the overall mean in percentage points

and indigenous women within the same schooling category, particularly for higher numbers of visits. The intersectionality analysis of schooling and country region also provides evidence for double jeopardy, with unschooled women residing in the North region having particularly low coverage levels (Supplementary Fig. 1).

Discussion

To interpret our findings, we rely on the inverse equity hypothesis, which posits that increased recommendations on the number of visits will initially reach the most affluent sectors of society, thereby potentially increasing inequality in coverage and health outcomes, exposing specific vulnerable groups of women [12]. This hypothesis will tend to have a greater effect in countries where market forces dominate access to healthcare provision, particularly where financial barriers related to user fees predominate. Nevertheless, the laws have also been shown to operate in countries with national health services, such as the United Kingdom, due to the maldistribution of resources, as well as cultural and informational barriers related to recognition of the importance of evidence-based care and to access to technology and digital literacy [19]. In Brazil, these factors contribute to the persistence of health inequalities [20] even though the implementation of the universal, publicly financed Unified Health System in 1990 contributed to marked improvements in maternal and child health [21].

Whereas national recommendations for four or more ANC visits were introduced in the early 2000s, advice on the need for six, seven, or eight visits is quite recent [7–9]. Thus, higher numbers of recommended visits may be characterized as new health interventions and might be expected to behave as predicted by the "inverse laws".

We found consistent results when analyzing the four stratification variables. First, the summary measures of inequality (slope indices and mean difference from the overall mean) increased monotonically with larger numbers of required visits. Second, the most vulnerable categories included women with no schooling and, to a lesser extent, those with less than 8 years of schooling; indigenous, followed by black and brown women; adolescents; and women living in the North region. For adolescents, substantial inequalities in the number of visits may be

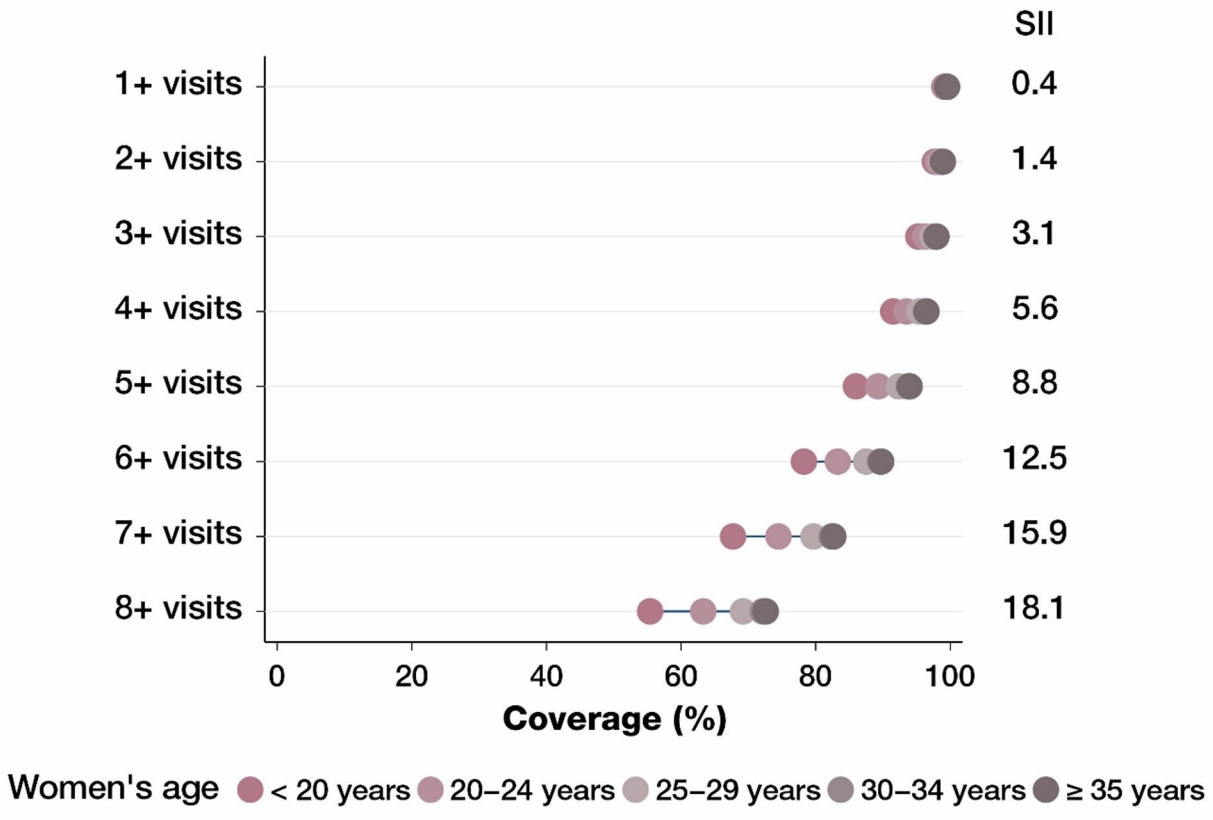

Fig. 4 Inequalities in the number of antenatal care visits according to women's age. SII: slope index of inequality in percentage points

related to delayed diagnosis of pregnancy, not allowing for more than four or five visits [3]. Lastly, these vulnerable groups had consistently lower coverage and higher dropout rates than the remaining groups of women studied, characterizing a pattern described as "bottom inequality" [22] or "marginal exclusion", where a majority of the population enjoys access to service networks, but substantial groups remain excluded [23].

We found that schooling-related inequalities were present within each category of race/skin color, and vice versa, indicating that different types of vulnerabilities are compounded when affecting access to antenatal care. Such analyses are useful for identifying the most vulnerable groups of a society, who often accumulate multiple characteristics, or cumulative disadvantages, associated with poor health [24].

Although the use of the term "race" to describe individuals has been criticized [24], the Brazilian information systems have long relied on self-declared race/skin color to unveil important health inequalities. This classification is used to guide national policies promoting the health of black [25] and indigenous peoples [26], being endorsed by civil society movements representing the black population in the country [27].

Our findings that inequality tends to increase as more demanding indicators of coverage are adopted are consistent with the earlier research showing increasing socioeconomic inequality gaps for vaccine coverage with one, two, three, or four of the basic vaccines recommended for infants [28].

We showed that the four stratifiers used in our analyses (maternal schooling, race, and age, and geographic region of the country) were strongly associated with the proportion of women reaching the recommended numbers of ANC visits. Given the essential role of ANC for reducing maternal, newborn, and child mortality globally, it is expected that the patterns of early adoption of multiple ANC visits will influence future mortality and as predicted by the inverse laws, may contribute to exacerbating inequities in health outcomes, at least in the short term.

The magnitude of absolute measures of inequality, including those presented in this study, is not independent of the overall prevalence of the outcome in the population. Other factors being equal, the size of absolute inequalities tends to be greatest when the prevalence of the outcome is approximately 50% [29]. When the prevalence is very high, such as the case for at least one antenatal care visit in our analyses, absolute inequalities

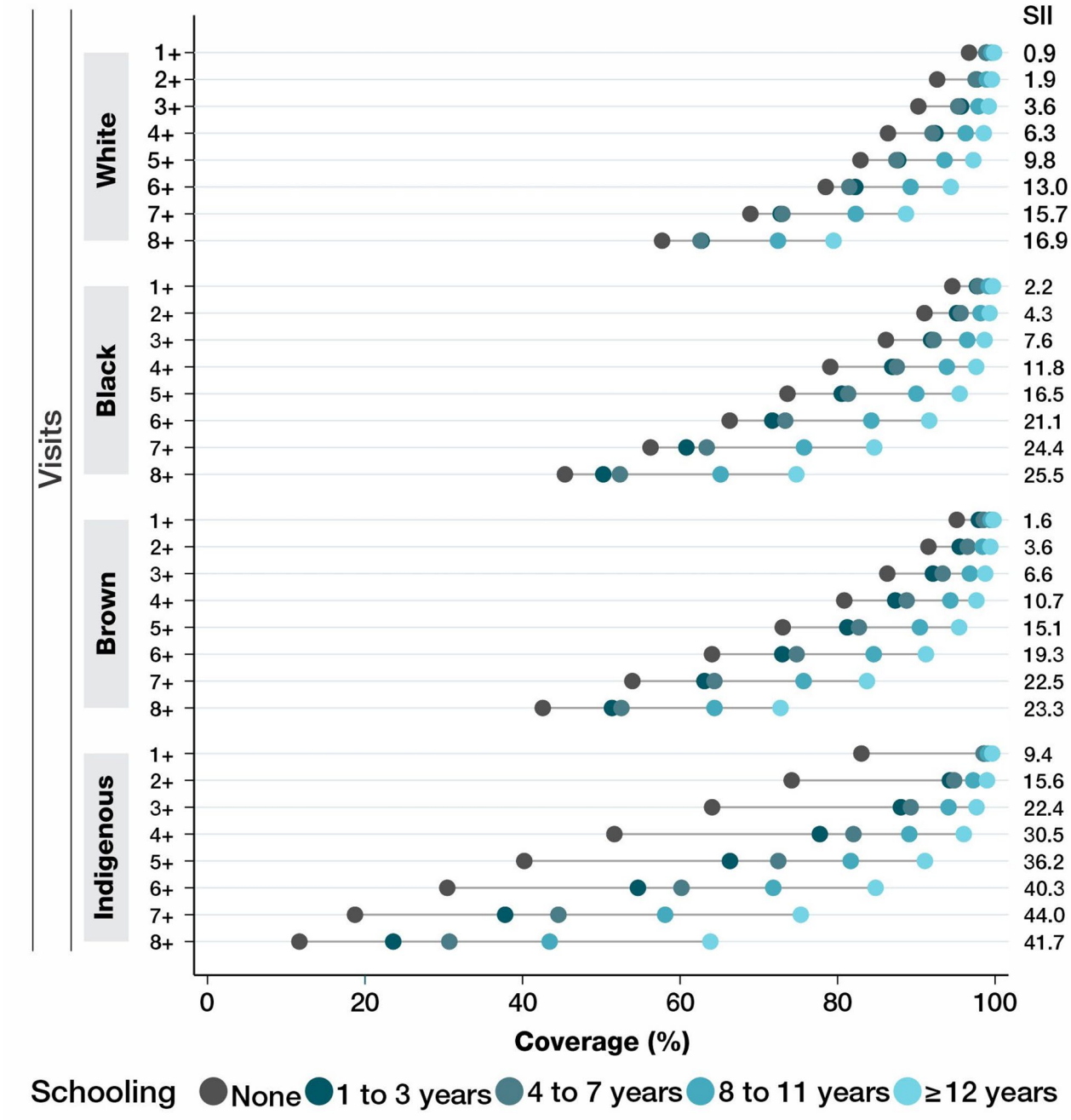

Fig. 5 Inequalities in the number of antenatal care visits according to women's schooling and race/skin color. SII: slope index of inequality in percentage points

are necessarily small due to a ceiling effect, as coverage approaches universality. Conversely, as prevalence declines, absolute inequality may initially increase but then tends to decrease once prevalence falls substantially below 50%. This is exemplified by the coverage of eight or more antenatal care visits in low-income countries, where the overall prevalence is close to zero; in such contexts, absolute inequality measures will be very small, as even the most privileged women are unlikely to achieve this coverage goal. In such countries, the levels of absolute inequality indices for 4 + visits (which typically show national coverage levels of around 70%) are larger than those for 8 + attendances (for which the average coverage is below 10%) [https://www.equidade.org/en/retriever].

The observed inequalities must be interpreted in light of what is known about the structural drivers of women's

and children's health in Brazil, with a historical, sociocultural, and political perspective [30–33]. Regarding regional inequalities, the North and Northeast are the poorest regions in the country, with less access to health services and human resources [33]. In addition, these regions are characterized by the presence of indigenous groups (particularly the North), early marriage and teenage pregnancy, all of which may explain the inequality patterns [34–36]. In terms of ethnic group inequalities, a nationwide survey of indigenous health documented late initiation of antenatal care that contributes to fewer visits, the survey also showed important limitations in the quality of antenatal care available to these women [36]. Black and brown women are also affected by systemic racism and discrimination in health care facilities [37, 38]. This accounts for their lower number of ANC visits than white women. Educational inequalities are also well documented in the country, a finding that is consistent with the inverse association between years of schooling and ANC coverage reported here [39]. Women with no schooling had particularly low coverage, but they accounted for only 0.6% of all deliveries. When interpreting inequalities by separate stratification variables, it is important to account for the intersectionality among the dimensions of vulnerability, as many women simultaneously belong to more than one group affected by social exclusion.

Our analyses have limitations. The first concerns the use of secondary data. SINASC is estimated to cover over 99% of all live births in the country, ranging from 97.8% in Roraima state which has a large indigenous population to 100.0% in Rio Grande do Sul, the southernmost state in the country. Despite high coverage, the data may be affected by omissions or errors in the Live Births Declaration forms completed in maternity hospitals nationwide under routine conditions [40]. Missing data may also have affected our results, but it is reassuring that no more than 2.5% of all records were affected. Stillbirths, which are not reported to the system, tend to be associated with shorter gestational ages than live births, and are also more frequent among women from vulnerable groups who are less likely to have multiple ANC visits [41–43]. Had stillbirths been included in the analyses, one might have expected even wider inequalities. An additional limitation is the lack of information on ANC quality, as SINASC only provides data on the number of visits, quality varies substantially with women's sociodemographic characteristics, and ANC quality inequalities are likely more pronounced than those reported here [44]. Lastly, the stratified analyses were restricted to variables available in SINASC, and other potential drivers of inequality, such as household wealth or income, could not be studied.

The present analyses were made possible due to substantial investments in national health information systems that have taken place since the SUS was launched over 30 years ago [11]. By also collecting information on education, race, and other determinants of health, the SINASC system is a key source of disaggregated information that may be used to assess the impact of governmental policies over time. For example, in the near future it will be possible to evaluate how inequalities in coverage with, for example, seven or more ANC visits, have evolved since this recommendation was proposed in 2024, using the present 2023 results as a baseline, thus allowing one to put the inverse equity hypothesis to further testing.

Conclusions

The adoption of higher ANC standards in Brazil reveals wider inequalities than were observed when using less ambitious recommendations for fewer visits. The higher cutoffs reveal the special vulnerability of indigenous, with no formal schooling, adolescents, and of those living in the North region. Policymakers should be made aware that newer, stricter guidelines for health coverage may disclose wider inequality gaps than those observed with earlier recommendations. The more precise detection of inequalities should prompt policymakers to focus on addressing barriers to ANC access and utilization. This will ensure that essential, high-standard care reaches all women and children, thereby avoiding the exacerbation of existing gaps in health outcomes, such as maternal and newborn mortality.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12939-026-02811-4.

Supplementary Material 1

Acknowledgements

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Author contributions

LESS participated in the study design, drafting of the results, interpretation of the statistical analyses and results, writing of the first draft of the manuscript, and critical review of the final version. CV conceived the article, participated in study planning, design, interpretation of statistical analyses and results, and was responsible for the final draft. JCC, LPVR, CB, and AJDB participated in the analyses and interpretation of the results, and critical review of the article. All authors approved the final version of the article, had full access to all the data in the study, and had final responsibility for the decision to submit for publication.

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Data availability
The datasets generated and/or analysed during the current study are available in the Ministry of Health repository, (https://datasus.saude.gov.br/transferencia-de-arquivos/).

Declarations

Ethics approval and consent to participate
All information on mothers and live births is anonymized, and the analyses do not require approval by a local ethics committee.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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